Adaptive Enhancement of Stuttered Speech Correction

Project Report

submitted by

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Reg. No. MAC21CD014

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Reg. No. MAC21CD036

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to

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

in partial fulfilment of the requirements for the award of the Degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Department of Computer Science & Engineering

Mar Athanasius College of Engineering (Autonomous)

Kothamangalam, Kerala, India 686 666

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DECLARATION

I undersigned hereby declare that the project report "Adaptive Enhancement of Stuttered Speech Correction", submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Prof. Leya Elizabeth Sunny. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Kothamangalam

Date:

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAR ATHANASIUS COLLEGE OF ENGINEERING

(AUTONOMOUS)

KOTHAMANGALAM

CERTIFICATE

This is to certify that the Project report entitled **Adaptive Enhancement of Stuttered Speech Correction** *submitted by* **Mr.Basil Anil (MAC21CD014), Ms.Melissa Pearson(MAC21CD036), Ms.Namya Sushil(MAC21CD043)**, *towards partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering (Data Science) from APJ Abdul Kalam Technological University for March 2025 is a bonafide record of the project carried out by them under our supervision and guidance.*

Prof. Anu Eldho Project Guide	**Prof. Kripa Joy** Project Guide
Prof. Leya Elizabeth Sunny Project Coordinator	**Prof. Joby George** Head of the Department
Internal Examiner(s)	External Examiner(s)
Date:	Dept. Seal

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ABSTRACT

To tackle the challenge of stuttered speech, a real-time, language-independent system has been developed to automatically correct common speech disfluencies, including repetitions, prolongations, and extended pauses, using digital signal processing techniques. Designed for high accuracy and efficiency, the system ensures practical applicability across multiple languages. A key aspect of this approach involves leveraging artificial intelligence and machine learning to dynamically adjust thresholds, allowing for adaptation to individual speaker variations and improved correction performance. Beyond stutter correction, the system also addresses other speech disorders, such as syllable repetition and interjections. Looking ahead, integration with popular voice assistants like Google Assistant, Siri, and Alexa is envisioned to enhance accessibility for individuals with speech disfluencies, providing a more seamless voice-enabled experience.

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List of Abbreviations

MFCC	Mel Frequency Cepstral Coefficients
LPC	Linear Predictive Coefficients
PLP	Perceptual Linear Prediction
DTW	Dynamic Time Warping
SVM	Support Vector Machine
TTS	Text-to-Speech
DSP	Digital Signal Processing
ANN	Artificial Neural Network
k-NN	k-Nearest Neighbors
SLP	Speech-Language Pathologist
VQ	Vector Quantization
LPCC	Linear Predictive Cepstral Coefficients
HMM	Hidden Markov Mode
RNN	Recurrent Neural Network
ASR	Automatic Speech Recognition
AI	Artificial Intelligence

Chapter 1

Introduction

The issue of stuttering, a common speech disorder impacting around 1% of the global population, presents significant communication challenges. Individuals with this condition often experience involuntary repetitions, prolonged sounds, and extended pauses in speech. These disfluencies interfere with smooth verbal expression and make day-to-day interactions challenging, especially in situations where speech fluency is critical, such as public speaking or phone conversations. Despite the prevalence of stuttering, voice recognition systems, including those used in popular devices, struggle to accommodate this group. The lack of inclusive technology in this area highlights the need for specialized approaches to facilitate more accessible and reliable communication for those affected by stuttering.[1]

Historically, various models and techniques have been applied to recognize stuttering patterns, using tools such as Artificial Neural Networks (ANN), Hidden Markov Models (HMM), and Support Vector Machines (SVM). However, approaches that aim to correct stuttered speech remain relatively underexplored, with most focus placed on recognition rather than correction. Signal processing techniques offer a computationally efficient alternative to these complex neural models, making them a promising avenue for stutter correction, particularly for real-time applications where computational simplicity is essential.

Correction of stuttering using signal processing leverages several specific methods tailored to address the main types of disfluencies. Techniques such as Mel Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coefficients (LPC) have shown potential in identifying and correcting repetitive or prolonged segments of speech. Short-time energy calculations, alongside cross-correlation analyses between consecutive frames, allow the system to pinpoint repetitions and prolongations effectively. This precise identification is crucial in ensuring that disfluent speech patterns are adjusted without compromising the natural flow of speech.

An effective algorithm for stutter correction not only needs to detect repetitions and prolongations but also manage long pauses that often disrupt speech continuity. By analyzing time gaps between words, the system can differentiate between natural pauses and those caused by stuttering, removing the latter while retaining necessary, brief pauses. The method uses a calculated threshold, eliminating excessive pauses while maintaining the rhythm of normal conversation. Such algorithms can also operate across multiple languages, enhancing their applicability and ensuring they remain effective regardless of linguistic differences.

The role of stutter correction in improving interaction with voice-enabled technology cannot be overstated. Voice assistants and other speech-based interfaces are now widely integrated into daily life, offering conveniences that have become nearly indispensable. However, people with stuttering disabilities often face limitations when using these systems, as traditional voice recognition algorithms struggle to interpret stuttered input accurately. An inclusive approach that corrects disfluencies opens the door for a broader population to use these devices effectively, fostering greater independence and accessibility.

In addition to enhancing personal device usability, stutter correction technology holds promise for broader applications, including support in educational, social, and professional settings where verbal communication plays a vital role. Speech therapy and language pathology, fields in which accurate stutter correction could significantly aid patient assessment and treatment, are also likely to benefit. By facilitating smoother verbal interactions, this technology provides a stepping stone towards improved communication confidence for individuals facing speech challenges.

The correction techniques developed in recent studies demonstrate notable success, achieving high accuracy rates across various stuttering patterns. Through continuous improvements, particularly by refining thresholds for better accuracy and expanding the approach to include additional types of speech impediments, the potential for enhanced stutter correction becomes even more promising. Integrating these solutions within current voice recognition systems on mobile and desktop platforms would represent a significant step towards inclusive technology, offering individuals with stuttering greater agency and freedom in communication.

Chapter 2
Literature Review

2.1 Automatic Detection of Prolongations and Repetitions using LPCC

This study focuses on identifying speech disfluencies, particularly prolongations and repetitions, in stuttered speech. The authors utilize Linear Predictive Cepstral Coefficients (LPCC) to extract meaningful features that capture vocal tract characteristics, distinguishing fluent from disfluent speech. The LPCC approach is computationally efficient, making it suitable for real-time applications where quick detection is essential. The methodology effectively pinpoints the spectral variations that mark stuttered speech segments, offering a reliable detection mechanism for prolongations and repetitions.

By targeting specific disfluency types, this model contributes to both automated detection and potential therapeutic tools. The LPCC-based system is proposed as an accessible solution for stutter detection, as it bypasses the complexity of more intensive machine learning methods. The paper's findings highlight the effectiveness of LPCC in identifying speech disfluencies with significant accuracy, suggesting further exploration of LPCC in real-time speech therapy applications for stuttering.

2.2 Theory and Applications of Digital Speech Processing

In their comprehensive work on digital speech processing, Rabiner and Schafer delve into foundational techniques and applications in speech processing. This book serves as a vital resource for understanding speech signal processing, including various feature extraction techniques that are applicable in stuttering detection. The authors discuss Linear Predictive Coding (LPC), Mel-Frequency Cepstral Coefficients (MFCC), and other techniques widely used in speech analysis, which are pivotal in capturing the nuances of speech necessary for identifying stuttered speech patterns.

For researchers focused on speech disorders, Rabiner and Schafer's work provides a detailed look at the algorithms that can be applied to the field of stutter detection. The principles outlined in this book aid in understanding how speech signals can be manipulated and analyzed, offering a theoretical foundation that has informed subsequent developments in automated stutter detection systems.

2.3 Vector Quantization and MFCC based Classification of Dysfluencies in Stuttered Speech

A method to eliminate silent pauses in stuttered speech using vector quantization (VQ) and Dynamic Time Warping (DTW). VQ provided data compression, while DTW was used to match disfluent speech samples with reference samples to identify and remove silent pauses. The model demonstrated success in reducing unwanted pauses, making it particularly relevant for real-time applications where excessive silence can disrupt communication.

VQ's ability to compress data helps reduce the overall computational load, making it useful in applications where resources are limited, such as mobile devices. By pairing VQ with DTW, the system achieves efficient silent pause identification, aligning speech samples with stored patterns to detect and correct unwanted pauses accurately. This approach was tested successfully across multiple languages, showing promise in multilingual applications.

However, DTW's high computational requirements, especially when paired with VQ, create challenges for time-sensitive tasks. DTW requires iterative comparisons to align samples, which can introduce delays and affect the model's performance in real-time applications. Additionally, while effective for silent pause removal, this approach is less adaptable to other types of stuttered speech corrections, such as repetition or prolongation, limiting its overall utility.

Our model takes a more lightweight approach, applying simple energy-based thresholding to detect and correct long pauses without requiring complex alignment algorithms like DTW. This approach allows faster processing times and reduces the dependency on reference samples, making it ideal for real-time applications where response speed is crucial. Additionally, by focusing on energy thresholds, we can retain natural pauses, enhancing the fluidity of corrected speech output.

Unlike the VQ and DTW combination, our model's use of correlation measures to detect dysfluency allows for broader applicability beyond silent pauses. By tuning thresholds based on empirical data, our model accurately identifies and corrects repetitions and prolongations, providing a more comprehensive solution for stutter correction. This expands our model's utility across various stutter types, making it versatile for diverse real-world applications.

Our threshold-based approach also proves more adaptable across different languages, as it does not rely on pre-defined reference samples. By dynamically adjusting thresholds, we maintain high performance across linguistic variations without the need for extensive retraining, enhancing usability for multilingual user bases. This adaptability also improves scalability, making our model suitable for deployment in global applications with minimal reconfiguration.

2.4 Feature Extraction and Signal Classification methods for Stuttering Speech Analysis

It review various feature extraction techniques like Cepstral Analysis, LPC, and MFCC for stuttering speech analysis, providing a framework for understanding their effectiveness in identifying stuttered speech. They compare classification algorithms and discuss their suitability for real-time stutter detection systems, highlighting the importance of selecting robust features for accurate classification.

Their findings indicate that MFCC and LPC feature extraction methods are highly effective in capturing the specific characteristics of stuttered speech. By evaluating signal classification techniques, this study offers insights into the feasibility of using such methods in stuttering therapy applications, laying the groundwork for developing accessible stutter detection tools.

2.5 Classification of Speech Dysfluencies Using Speech Parameterization Techniques

This study examines the classification of speech dysfluencies using parameterization techniques, focusing on MFCC, LPC, and LPCC for feature extraction. By employing these methods, the authors aim to accurately identify types of dysfluencies in speech, including syllable and word repetitions as well as prolongations, achieving significant classification accuracy.

The research emphasizes the advantages of using LPCC and other spectral features for capturing the subtleties of disfluent speech, which are often missed by simpler algorithms. This work demonstrates the practical applicability of these techniques in building automated tools that could be used in real-time stuttering therapy.

2.6 Automatic Correction of Stutter in Disfluent Speech

Speech processing and stuttering correction have been active areas of research for several decades. Early works in speech processing focused on fundamental techniques such as short-time energy analysis, pitch detection, and frequency-domain transformations to analyze and modify speech signals (Rabiner Schafer, 1978). The development of feature extraction techniques like Mel-Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC) further improved speech analysis, enabling better recognition and classification of speech patterns (Davis Mermelstein, 1980). These foundational studies laid the groundwork for subsequent advancements in speech signal processing, including stutter correction methodologies.

The issue of stuttering and its automatic correction has been extensively studied in the field of speech pathology and computational linguistics. Early research focused on manual intervention techniques such as delayed auditory feedback and rhythm training (Bloodstein Ratner, 2008). However, with the advent of digital signal processing, researchers began exploring algorithmic approaches to identify and correct stuttered speech. One such approach involves detecting and eliminating repeated phonemes and words using statistical pattern recognition methods (Howell et al., 2011). These studies demonstrated that computational methods could significantly enhance speech fluency without the need for intensive human intervention.

Recent advancements in machine learning and deep learning have further enhanced speech processing applications. Researchers have employed artificial neural networks, hidden Markov models, and deep recurrent networks to detect speech disfluencies, including repetitions, prolongations, and pauses (Ravanelli et al., 2019). MFCC and LPC features have been widely used in these studies to analyze speech characteristics and distinguish between normal and disfluent speech patterns. Studies have shown that MFCC features tend to outperform LPC features in various speech processing tasks due to their robustness in capturing spectral properties (Young et al., 1997). This aligns with the findings in the present study, where MFCC-based analysis achieved higher accuracy than LPC-based methods.

Several studies have also focused on pause analysis and its impact on speech fluency. Researchers have investigated the significance of pause duration in natural speech and stuttered speech, emphasizing that long pauses disrupt fluency and hinder effective communication (Katahira et al., 2018). Algorithms designed to shorten unnatural pauses while retaining necessary ones have shown promise in improving speech fluency. Previous works have experimented with different threshold values for pause duration, optimizing speech naturalness while ensuring computational efficiency. The threshold-based approach employed in this study aligns with earlier research findings, demonstrating effective pause correction.

2.7 Speech Disorder Recognition using MFCC

This research utilized a Support Vector Machine (SVM) classifier combined with MFCC feature extraction for dysfluency detection, achieving high accuracy rates for classifying syllable repetition, word repetition, and prolongation. SVM's robust classification ability allows it to handle complex patterns effectively, making it well-suited for distinguishing between various types of speech dysfluencies. The study reported notable success, with 76.%, 88%, and

92% accuracy for different dysfluency categories.

The SVM's ability to handle high-dimensional data makes it advantageous for applications requiring intricate pattern recognition, such as stuttered speech classification. By incorporating MFCC, the model captures spectral features essential for distinguishing dysfluent speech patterns, contributing to high classification accuracy. This accuracy in classifying disfluencies has practical implications in speech therapy, where accurate dysfluency categorization is essential for developing effective treatment plans.

However, SVM's limitations become evident when applied to real-time speech correction, as its high computational demands and slower processing times affect performance in time-critical applications. SVM also requires extensive training for each dysfluency type, reducing flexibility when adapting to new languages or accents. This makes it challenging for applications requiring rapid adaptability, such as mobile voice assistants or real-time communication aids.

To overcome these challenges, our model substitutes SVM with a threshold-based method that applies real-time correlation and energy analysis for dysfluency detection and correction. By avoiding complex classification algorithms, we reduce computational overhead and achieve faster processing, enabling our model to function effectively in real-time settings. This threshold-based approach maintains accuracy while allowing immediate correction, essential for seamless user experiences in voice-enabled devices.

Moreover, unlike SVM, which may struggle with language adaptability, our model's use of generalizable thresholds allows it to function consistently across multiple languages without requiring retraining. By tuning the thresholds based on experimental data, our model achieves dynamic adjustments to various speaking rates and contexts, ensuring accuracy across diverse environments. This adaptability significantly enhances our model's suitability for applications with broad linguistic diversity.

Our system also expands beyond classification, implementing direct correction of disfluencies by removing prolonged or repeated segments in real time. This goes beyond the detection focus of the SVM-based approach, offering practical benefits for real-world users who rely on immediate stutter correction. The efficient processing capabilities of our threshold-based method also reduce latency, enhancing user experience in real-time applications.

2.8 An Automatic Prolongation Detection Approach in Continuous Speech with Robustness Against Speaking Rate Variations

This paper introduces a method for detecting prolongations in continuous speech that is robust against variations in speaking rate. By utilizing Perceptual Linear Predictive (PLP) features, the model effectively identifies prolonged sounds, even when speech rate fluctuates significantly. The model achieved 99% accuracy on the UCLASS database and 97.1% on the Persian database, showing strong reliability across languages.

The advantage of PLP features lies in their perceptual basis, which captures essential characteristics of speech as perceived by human listeners. This makes PLP highly suitable for applications where natural-sounding corrections are essential, as it closely aligns with human auditory processing. By effectively handling speaker rate variations, the model demonstrates its robustness and adaptability, providing reliable stutter detection even when speech speedfluctuates. However, the reliance on PLP features introduces certain limitations.PLP-based models are computationally demanding, which can impact real-time correction performance. Additionally, while effective for detection, the model does not directly address stutter correction, which limits its applicability in real-time environments where immediate correction is needed. The PLP approach also requires extensive configuration, which may not be feasible in multilingual or cross-cultural applications due to variations in speech patterns across languages.

2.9 Automatic Speech Recognition System for Stuttering Disabled Persons

Explored an automatic speech recognition system designed to assist stuttered speech detection by combining MFCC and LPCC features. This dual feature extraction approach captures a wider range of spectral information, helping the model distinguish between fluent and dysfluent speech segments. Their model proved effective in identifying stuttered speech with reasonable accuracy, offering potential applications in assistive technology for individuals with speech disorders.

Combining MFCC and LPCC allows the model to capture both the detailed spectral characteristics (MFCC) and the predictive coding benefits (LPCC) of speech. This combination enhances the accuracy of stutter detection by providing a broader feature set for classification. The model's ability to differentiate between fluent and dysfluent speech makes it valuable for use in clinical settings, where accurate stutter identification is essential for therapy and treatment.

However, the use of two feature extraction methods increases computational demand, posing challenges for real-time applications where processing time is limited. Additionally, the model's detection-only focus limits its applicability in real-world environments, where immediate correction is often required. Without direct correction capabilities, the system may be less suitable for interactive applications like voice-enabled devices.

Our model streamlines the feature extraction process by utilizing MFCC and LPC alone, maintaining a balance between feature richness and computational efficiency. By focusing on real-time correction, our model not only detects stuttered speech but also corrects it by removing prolonged and repeated segments. This correction capability enhances our model's practical applications, providing a seamless experience in interactive environments where immediate response is essential.

In place of complex feature combinations, our model uses adaptive thresholds to dynamically adjust based on speech patterns, making it more versatile across varied speech environments. This adaptability improves real-time accuracy without the need for multiple feature sets, reducing computational load while maintaining high performance. Our approach thus provides a more streamlined, efficient solution for real-time stutter correction.

Our model's thresholding method is also language-independent, enabling consistent performance across multiple languages without requiring specialized configurations. By focusing on universal signal characteristics, we ensure that our model functions effectively in multilingual contexts, making it highly adaptable for global applications. This versatility addresses a key limitation in the MFCC-LPCC model, which may require language-specific tuning for optimal performance.

The MFCC-LPCC approach provides accurate stutter detection, our model's threshold-based corrections and streamlined feature extraction offer a more practical solution for real-time, multilingual applications, enhancing user experience in interactive speech environments.

2.10 Utilizing Perceptual Linear Predictive (PLP) Features in Stutter Detection

Developed a model that focused on using Perceptual Linear Predictive (PLP) features to enhance the accuracy of stutter detection, especially in scenarios with varied speaking rates. PLP features, which model human auditory perception, improve accuracy by capturing perceptually relevant information. The model used these features to distinguish between fluent and dysfluent speech, achieving an impressive accuracy rate of 98.05% across English and Persian language datasets, making it robust against speaker rate variations.

The advantage of PLP features lies in their perceptual basis, which captures essential characteristics of speech as perceived by human listeners. This makes PLP highly suitable for applications where natural-sounding corrections are essential, as it closely aligns with human auditory processing. By effectively handling speaker rate variations, the model demonstrates its robustness and adaptability, providing reliable stutter detection even when speech speed fluctuates.

However, the reliance on PLP features introduces certain limitations. PLP-based models are computationally demanding, which can impact real-time correction performance. Additionally, while effective for detection, the model does not directly address stutter correction, which limits its applicability in real-time environments where immediate correction is needed. The PLP approach also requires extensive configuration, which may not be feasible in multilingual or cross-cultural applications due to variations in speech patterns across languages.

Our model overcomes these challenges by focusing on MFCC and LPC features, which are computationally lighter and adaptable across diverse linguistic contexts. We incorporate adaptive thresholds that dynamically adjust based on real-time speech characteristics, enabling consistent performance across varied speech speeds without complex configurations. This approach simplifies our model, enhancing its usability in real-time applications where rapid detection and correction are critical.

By leveraging signal processing techniques rather than perceptual models like

PLP, our model maintains accuracy with a significantly reduced computational load, supporting real-time applications in mobile and embedded devices. This computational efficiency is crucial for live voice interactions, where immediate processing is essential for smooth communication. Our adaptive thresholding also ensures that speech corrections are accurately applied regardless of speech rate, eliminating the need for extensive speaker rate adjustments.

Our model's cross-linguistic adaptability further enhances its applicability, as it does not require language-specific tuning. This universality improves usability in diverse environments, allowing the model to be seamlessly integrated into global applications. This is particularly important in speech correction applications designed for users across various language backgrounds, where consistency in correction performance is key.

Therefore PLP offers high accuracy for detection, our model's simpler, adaptive approach provides a comprehensive solution for real-time stutter correction. Our focus on dynamic thresholding and lightweight feature extraction ensures that the model is efficient, versatile, and suitable for multilingual applications.

Chapter 3
Objectives

The primary objective of this paper is to develop a robust, efficient, and real-time stutter correction system for disfluent speech, which effectively addresses common types of stuttering (prolongations, and long pauses) using signal processing techniques. This system aims to operate independently of language constraints and to function seamlessly across various devices, including low-power environments like mobile phones or embedded systems.

- **Addressing the Need for Real-Time Stutter Correction :** Stuttering affects approximately 1% of the global population, with adverse effects on communication and quality of life. With advancements in voice-enabled technology, the demand for real-time stutter correction systems has grown significantly. Current voice assistants, such as Siri, Alexa, and Google Assistant, are often inaccessible to individuals with stuttering disorders due to their lack of disfluency recognition. The objective of this paper is to bridge this gap by creating a real-time stutter correction system that is both effective in identifying disfluencies and capable of processing corrections in milliseconds, enabling smoother interactions with voice-enabled devices for individuals affected by stuttering.

- **Building an Efficient and Lightweight Model :** While many stutter correction methods rely on complex algorithms like neural networks or support vector machines (SVMs), these approaches are often computationally intensive and unsuitable for real-time applications. This paper aims to introduce a more lightweight approach using signal processing techniques, specifically Mel Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coefficients (LPC), to achieve efficient feature extraction. The system uses short-time energy and correlation measures, providing a simpler yet effective alternative to high-computation models. This allows the system to be implemented in low-power and real-time environments, making it more accessible and adaptable for applications on mobile or embedded devices.

- **Achieving Language Independence :** One of the challenges in stutter correction is ensuring that the model can function across multiple languages without requiring language-specific training data. Many existing models are limited to certain linguistic contexts, necessitating retraining or configuration changes for each language. The objective of this paper is to design a system that operates independently of linguistic differences. By focusing on universally applicable signal processing features, such as MFCC and LPC, the model aims to capture essential aspects of speech across languages. This language-independent design enhances the system's usability in diverse linguistic and cultural contexts, allowing individuals worldwide to benefit from stutter correction technology.

- **Improving Accuracy in Identifying and Correcting Disfluencies :** It seeks to improve upon previous systems by achieving high accuracy in both detecting and correcting three primary types of stutter: repetitions, prolongations, and long pauses. By experimenting with different thresholding techniques and tuning parameters, the model aims to detect disfluencies precisely and apply corrections without over-correcting or distorting the natural flow of speech. The system's accuracy goal is to match or exceed the performance of more computationally intensive models, thereby setting a new standard for accuracy in lightweight stutter correction systems.

- **Maintaining Naturalness in Corrected Speech :** An important aspect of stutter correction is preserving the natural flow and rhythm of speech. Systems that apply aggressive or overly stringent corrections risk altering the user's intended speech patterns, leading to unnatural or roboticsounding output. The objective of this paper is to design a correction process that retains essential speech characteristics, preserving the user's natural voice as closely as possible. This is achieved through fine-tuning parameters to avoid eliminating non-repeated words or essential pauses, which would otherwise disrupt the natural pacing of the corrected speech. By focusing on naturalness, the paper aims to create a system that supports fluent, organic-sounding speech post-correction.

- **Providing Adaptability Across User Profiles :** Speech disfluencies can vary widely between individuals, with differences in severity, frequency, and style of stutter. The objective of this paper is to design a system that adapts dynamically to these variations, ensuring that it provides effective correction across different user profiles. By implementing adaptive thresholding based on real-time speech analysis, the model adjusts to the unique disfluency patterns of each user, allowing it to cater to a broad spectrum of stuttering behaviors. This adaptability also means the system does not need extensive customization or retraining, making it more flexible for general use.

- **Enabling Seamless Integration with Existing Voice Technologies:** Aims to design a model that can be integrated smoothly with current voice recognition technologies. By operating independently of specific voice assistant systems, this model can serve as a universal plugin or add-on, enhancing accessibility for stutter-affected individuals across different platforms. This universal design philosophy allows for easy integration into various software and hardware configurations, from smart devices to desktop applications. The ultimate objective is for this system to become a widely accessible tool, enabling individuals with stuttering disorders to interact seamlessly with the latest in voice-enabled technology.

Chapter 4
Preliminary Analysis

In the preliminary analysis of Adaptive Enhancement of Stuttered Speech Correction existing methods and techniques used for stutter detection and correction, focusing on their performance, limitations, and applicability to real-time speech systems. This involves a review of both traditional and more recent computational approaches, such as neural networks, support vector machines (SVMs), and signal processing techniques. A notable observation from this analysis is that while many methods demonstrate high accuracy in detecting stuttered speech, they often fall short when it comes to real-time correction. For example, neural networks and SVM-based methods require extensive processing power, making them unsuitable for time-sensitive applications. This review helped shape the direction of the paper by highlighting the need for a computationally efficient and lightweight system that could operate in real-time environments.

The analysis further explores the feature extraction methods used in previous studies, such as Mel Frequency Cepstral Coefficients (MFCC), Linear Predictive Coding (LPC), and Perceptual Linear Predictive (PLP) features. These techniques have been employed in various ways to capture essential characteristics of stuttered speech, such as pitch, rhythm, and amplitude. MFCC and LPC, in particular, are frequently used due to their ability to capture the spectral properties of speech effectively and computationally efficiently. The paper's authors found that while PLP features provide high accuracy in stutter detection, they are computationally intensive, which limits their use in real-time applications. As a result, MFCC and LPC were selected as primary features for the proposed model due to their balance between accuracy and processing efficiency.

The preliminary analysis also examines different types of stuttering—repetitions, prolongations, and long pauses—and identifies the signal characteristics associated with each. For instance, repetitions often involve repeated word or syllable patterns, which can be detected through short-time energy calculations. Prolongations are characterized by extended sounds or syllables, which can be identified by analyzing correlation patterns between speech frames. Long pauses are marked by extended silence, distinguishable by a drop in signal energy. This understanding of the distinct characteristics of each disfluency type informed the development of specific correction algorithms tailored to address each stutter type individually. By focusing on signal characteristics that are unique to each stutter, the model can detect and correct stuttering with a higher degree of precision.

Another focus of the preliminary analysis was understanding the adaptability of various stutter correction systems across languages and user profiles. Many existing models are designed for specific languages or linguistic characteristics, making them less applicable in multilingual contexts. The analysis showed that language-independent signal processing features, such as MFCC and LPC, hold promise for creating a universal stutter correction model. By choosing features that do not rely on language-specific data, the model can be applied to a wide range of linguistic backgrounds, enhancing its usability for diverse user groups. This flexibility is crucial for real-world applications, where users may speak different languages or have unique speech characteristics.

The analysis includes a study of computational requirements and limitations of previous methods. High computational complexity in neural network-based models, for instance, makes them unsuitable for low-power devices and embedded systems, such as mobile phones or voice-assisted devices. The authors noted that approaches using simpler, threshold-based methods for feature extraction and analysis, such as short-time energy and correlation measures, offer a significant advantage in terms of processing speed and efficiency. This insight directed the research towards developing a threshold-based model that uses experimentally determined values for stutter correction. By focusing on computationally efficient methods, the authors aim to create a model that is accessible and implementable in a wide range of real-time applications, from desktop software to portable devices.

Analysis provided a clear foundation for the paper's objectives by identifying limitations in existing models, selecting efficient feature extraction techniques, tailoring algorithms to each type of stutter, ensuring language independence, and prioritizing computational efficiency. This background informed the design of a model that meets the practical needs of real-time stutter correction across diverse applications and user environments.

Chapter 5

Existing System

The existing systems for stutter detection and correction are broadly categorized into computational models that use machine learning, signal processing, and hybrid approaches combining neural networks and rule-based methods. While these systems have made significant strides in detecting speech dysfluencies, their ability to correct stuttered speech effectively and in real-time remains limited. Machine learning-based models, such as those utilizing support vector machines (SVM) and neural networks, excel at classifying stutter types with high accuracy. However, these models tend to have high computational requirements, which limits their use in real-time applications, especially on low-power devices. The analysis in the paper emphasizes that while these models are robust for offline stutter detection, they lack the necessary efficiency for real-time correction tasks, which require instantaneous processing to provide fluent speech output.

Signal processing techniques, on the other hand, offer a more lightweight approach and have been applied to feature extraction for stutter detection and correction. Mel Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC) are two popular feature extraction methods frequently used in these systems due to their efficiency in capturing spectral features relevant to stuttered speech. MFCC focuses on representing the power spectrum of speech, making it sensitive to changes in pitch and tone associated with speech dysfluencies. LPC provides an effective means of compressing speech data while retaining essential acoustic information. Although signal processing approaches provide computational efficiency, they often fall short of providing the sophisticated pattern recognition capabilities that machine learning models offer, particularly in complex speech environments.

In addition to standalone signal processing and machine learning models, hybrid systems have been developed to combine the strengths of both approaches. For instance, some models use neural networks for detecting stuttered speech patterns and then apply rule-based methods, such as thresholding, to perform correction. Another hybrid approach includes the use of Text-to-Speech (TTS) conversion, where detected disfluencies are removed, and the corrected text is synthesized into fluent speech output. These systems can deliver accurate correction, but they are computationally intensive and may produce unnatural sounding speech due to the synthetic nature of TTS outputs. Furthermore, such hybrid systems generally require a large amount of training data, limiting their adaptability to diverse linguistic backgrounds and real-time applications, where speed and naturalness are essential.

One significant limitation in many existing stutter correction systems is their language dependence. Several models rely on language-specific datasets and features, meaning they require retraining or configuration adjustments to function effectively in different linguistic contexts. This restricts their practical application, especially in multilingual environments where users may expect universal functionality regardless of their language. Systems that use perceptual linear predictive (PLP) features, for example, are tuned to specific auditory characteristics that may vary across languages, making them less adaptable. Consequently, language-dependent models are not suitable for broad, real-world applications that need to serve users across various linguistic and cultural backgrounds. The paper highlights the importance of a language-independent approach that can accommodate diverse speech patterns without extensive customization.

Lastly, the existing systems are often challenged by the need to balance accuracy with computational efficiency. While machine learning models can achieve high accuracy, they require extensive computational resources, which limits their integration into real-time applications such as mobile or embedded systems. Signal processing approaches, although computationally efficient, may not reach the same level of accuracy in detecting complex speech dysfluencies like repetitions and prolongations. This trade-off between accuracy and efficiency has led to a gap in current stutter correction solutions, with no single system addressing both needs comprehensively. The paper's review of these existing systems underscores the need for a solution that combines the simplicity and speed of signal processing with the adaptability and accuracy of more advanced models, aiming for an optimized system that can handle real-time stutter correction effectively.

Chapter 6

Proposed Systems

The proposed system processes speech data to enhance clarity by removing long pauses, and prolongations. It begins with the long pause removal stage calculates time differences between segments to distinguish natural pauses from overly long ones. Pauses exceeding a set time threshold are discarded to maintain fluidity. Finally, the prolongation removal step breaks speech into frames, then analyzes feature correlations within these frames to detect unnaturally extended sounds. Frames showing high similarity, indicating prolonged speech elements, are removed. Together, these steps refine the audio by filtering out redundancies and irregularities, leading to smoother, more natural-sounding speech output.

6.1 Prolongation Removal

1. Prolongation Removal Start:

    - This step initiates the process of identifying and eliminating prolonged sounds in the speech data. Prolongations are characterized by unnaturally extended sounds within words or between words, which can disrupt the natural flow of speech. The purpose of this process is to filter out these prolonged sounds to achieve a more concise and fluid audio output.
    - By starting this process, the system sets up an analysis sequence focused on detecting these elongated sounds. This setup involves prepar-

ing the speech data for further segmentation and feature analysis, ensuring that only prolonged elements are identified and removed in the following steps.

2. Divide into Frames:
   - In this step, the speech data is divided into frames, which are small segments of the audio signal. Breaking the data into frames allows the system to analyze short, manageable portions of audio, making it easier to detect subtle prolongations. Each frame is typically a fraction of a second long and contains a small portion of the audio that can be examined in isolation.
   - Dividing into frames enables detailed processing and increases the precision of the prolongation detection. By working with frames, the system can identify prolonged sounds at a granular level, comparing each frame's characteristics to detect patterns that indicate elongation.
3. Feature Extraction for Frames:
   - Once the audio is divided into frames, the system extracts key features from each frame. These features can include properties such as pitch, energy, frequency spectrum, and amplitude, which help characterize the audio content within each frame. Feature extraction is crucial because it provides data points that will later be used to determine whether a frame contains prolonged sound.
   - This extracted information helps the system create a profile of each frame's acoustic properties, which are essential for the next steps. By analyzing these features, the system can distinguish between normal speech sounds and prolonged sounds, as prolongations tend to have specific patterns, such as sustained pitch or amplitude.
4. Correlation Calculation of Frames:
   - After extracting features, the system calculates the correlation between frames. This calculation helps determine the similarity between

consecutive frames, identifying frames that may represent a prolonged sound. High correlation between adjacent frames often indicates that the sound has been extended, as prolonged sounds typically have consistent acoustic properties over time.

- By comparing the frames, the system can identify patterns of similarity that suggest elongation. If a series of frames shows high correlation, it implies that the same sound is persisting without variation, which is characteristic of prolonged speech elements. This correlation calculation is essential for detecting and flagging potential prolongations for removal.

5. Check Lengthy Correlation:

    - In this step, the system checks the calculated correlation values against a predefined threshold (in this case, a correlation threshold of 14). If the correlation between frames exceeds this threshold, it indicates that the frames are likely part of a prolonged sound. This step serves as a filtering mechanism to separate normal frames from those that indicate elongation.
    - By applying this threshold, the system ensures that only frames with excessive correlation (high similarity over time) are flagged as prolonged. Frames that do not meet this threshold are considered to contain normal speech and are allowed to pass through without further modification.

6. Eliminate Prolonged Frames (Above 14):

    - Frames with a correlation score above the threshold (14) are identified as prolonged and are eliminated from the output. Removing these frames ensures that the final audio output is free from unnecessarily elongated sounds, enhancing the naturalness of the speech. This elimination step reduces redundancy and prevents the audio from sounding drawn out or repetitive.
    - By discarding these frames, the system improves the speech quality,

making it sound more concise and less monotonous. This step is crucial for applications where clear, uninterrupted speech is essential, such as in virtual assistants or automated customer service.

7. Retain Initial Frames (Below 14):

   - If the correlation score is below the threshold, the frames are considered to be part of natural speech and are retained in the final output. Retaining these frames ensures that the system preserves the integrity of the original speech, only removing elements that are confirmed as prolonged. This selective approach helps maintain a balance between clarity and naturalness.
   - By keeping these initial frames, the system prevents the audio from being over-processed or sounding artificial. This step is essential for achieving a realistic and fluid final audio output, as it ensures that only unnecessary prolongations are removed while preserving the original speech flow.

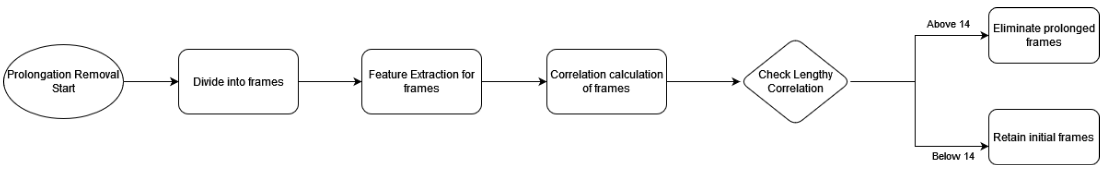


Figure 6.1: Prolongation Flowchart

6.2 Long Pause Removal

1. Long Pause Removal Start: This step marks the beginning of the long pause removal process. Here, the system prepares to analyze the audio data specifically to detect and eliminate excessively long pauses. Long pauses in speech can interrupt the flow and make the speech sound unnatural, so this step aims to smoothen the speech by identifying and removing these pauses while preserving natural, brief pauses.

2. Calculate Time Difference Between Words: In this step, the system calculates the time intervals between consecutive words or speech segments.

This involves analyzing the audio signal to measure the duration of silence between spoken words. By determining these time gaps, the system can identify where pauses occur and quantify their lengths. This time difference calculation is essential for distinguishing between natural short pauses and longer, potentially unnecessary pauses.

3. Check Time Threshold: Once the time differences are calculated, the system compares each pause duration against a predefined threshold of 0.5 seconds. This threshold serves as a boundary to separate natural pauses from long pauses. If the time difference between words exceeds 0.5 seconds, it is flagged as a long pause; otherwise, it is considered a natural pause. This threshold-based approach allows the system to consistently identify pauses that may need removal without affecting the natural rhythm of speech.

4. Eliminate Long Pause Samples (Above 0.5s): If a pause duration is above the 0.5-second threshold, it is classified as a long pause, and the system proceeds to eliminate it from the audio. Removing these long pauses helps make the speech sound smoother and more continuous, enhancing the overall listening experience. By eliminating unnecessary gaps, the system produces a more concise and engaging output, which is particularly beneficial in applications like automated voice responses.

5. Retain Natural Pause Samples (Below 0.5s): If the pause duration is below the 0.5-second threshold, it is deemed a natural pause and retained in the final audio output. Natural pauses contribute to the speech's expressiveness and help convey the intended meaning, so it's essential to preserve them. Retaining these brief pauses allows the processed audio to maintain its natural rhythm and flow, ensuring that the speech remains clear and understandable while only removing disruptions caused by overly long pauses.

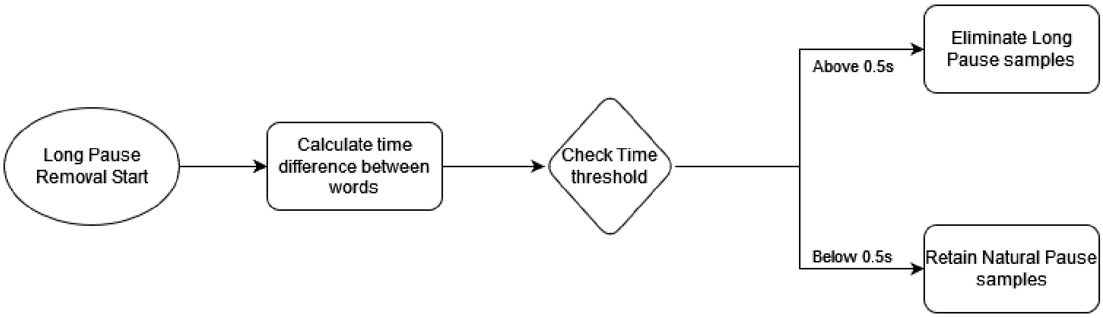


Figure 6.2: Long Pause Flowchart

6.3 Adaptive Learning

1. The meta-learning process is central to refining the system's ability to handle stuttering. Using the Reptile MAML (Model-Agnostic Meta-Learning) approach, the system adapts its processing parameters over 10 fixed iterations. Each iteration allows the model to learn from the previous processing steps and adjust its internal parameters to optimize the stutter reduction. The core of this learning is based on comparing the Mel-Frequency Cepstral Coefficients (MFCCs) of the processed audio with the original audio. The difference between these MFCCs is used to compute a stutter score, given by the equation:

$$\text{Score} = e^{-\text{mean}(|\text{MFCC}_{\text{orig}} - \text{MFCC}_{\text{proc}}|)} \tag{6.1}$$

2. This score quantifies how similar the processed audio is to the original, with higher values indicating that the speech's essential characteristics are better preserved during processing. This comparison helps ensure that stutter reduction doesn't result in unnatural speech distortions, maintaining speech quality while reducing stuttering.

3. Once the stutter score is calculated, the system updates its internal thresholds—such as those for length, energy envelope, and correlation—via a meta-update. This is done using the gradient descent update rule:

$$\Delta\theta = -\alpha \nabla L \tag{6.2}$$

4. where θ are thresholds, α is the learning rate, and $L = 1 - \text{Score}$ is the loss function. This loss function represents how much the processed audio deviates from the original, guiding the updates to improve future iterations. After each update, the system tracks the scores and keeps the best iteration, which corresponds to the optimal performance for that specific recording. The system's real-time feedback mechanism enhances its adaptability. After each iteration, the processed audio is immediately played back to the user, allowing them to hear the effects of the changes in real-time. This iterative process runs for 10 iterations per recording, progressively refining the audio's quality by adjusting the thresholds based on the outcomes of the meta-learning process. Users can also interrupt this loop at any time by pressing the 'q' key, ensuring they have control over the process.

5. At the conclusion of the 10 iterations, the system reports the final thresholds and the best stutter reduction score achieved. This structure allows for multiple recordings, with each new recording contributing to the ongoing refinement of the system's parameters. By learning from each session, the system becomes more effective at stutter reduction over time, making it a powerful tool for speech therapy or real-time speech enhancement applications. This iterative learning process ensures continuous improvement, adjusting to each user's unique speech patterns for optimal fluency enhancement.

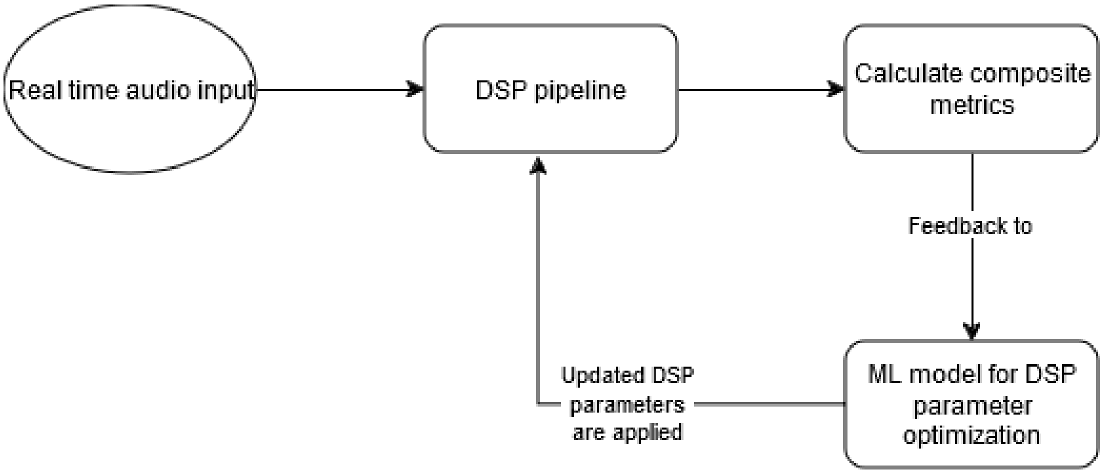


Figure 6.3: ML model overview

Chapter 7

Results

The outcomes presented here are derived from our experi- mental investigation, showcasing 10 iterations on a stuttered audio file before and after stutter removal. In order to simulate adaptive stutter correction over time, we performed multiple iterations on the same audiofile. The aim is that in real time, with multiple stuttered audios from a single person, the model will adapt to their speaking style.

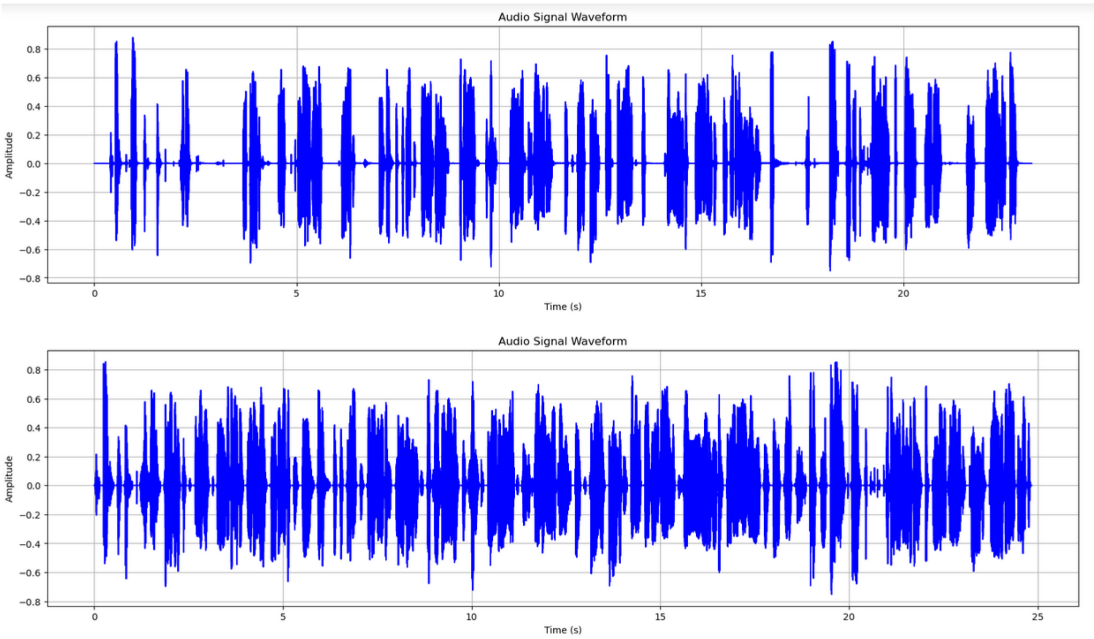


Figure 7.1: First Iteration

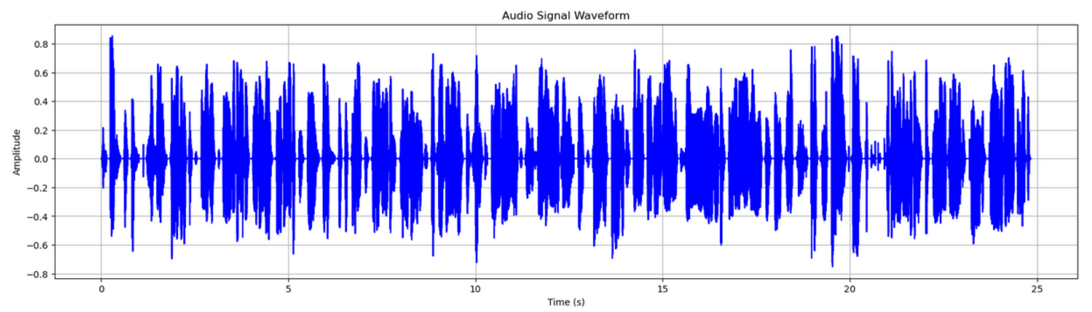


Figure 7.2: Second iteration

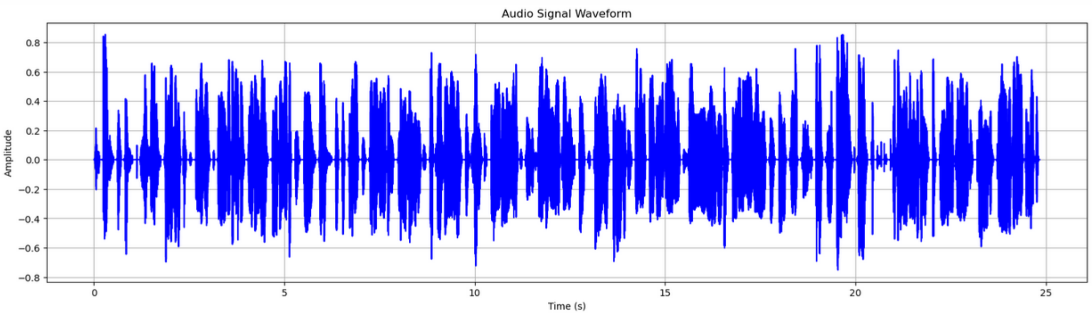


Figure 7.3: Third iteration

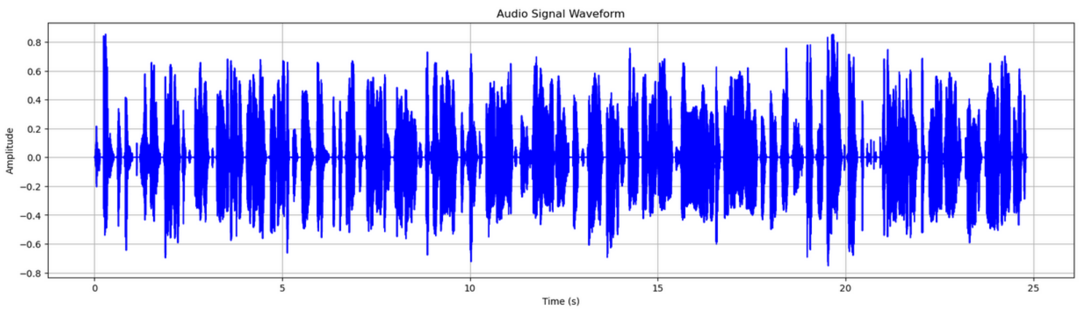


Figure 7.4: Fourth iteration

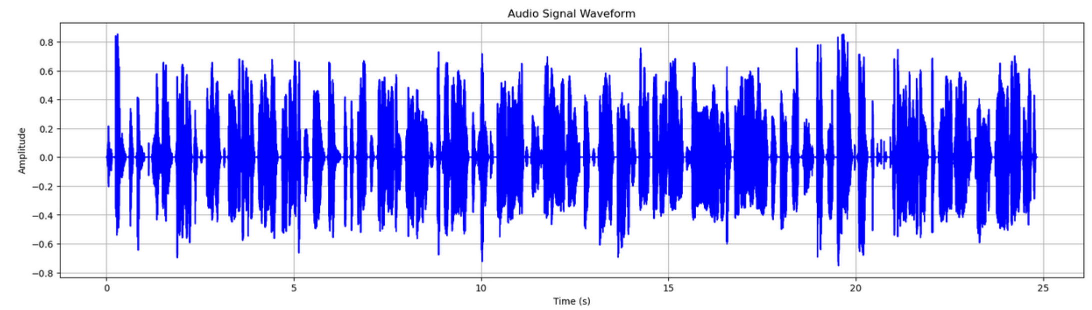


Figure 7.5: Fifth iteration

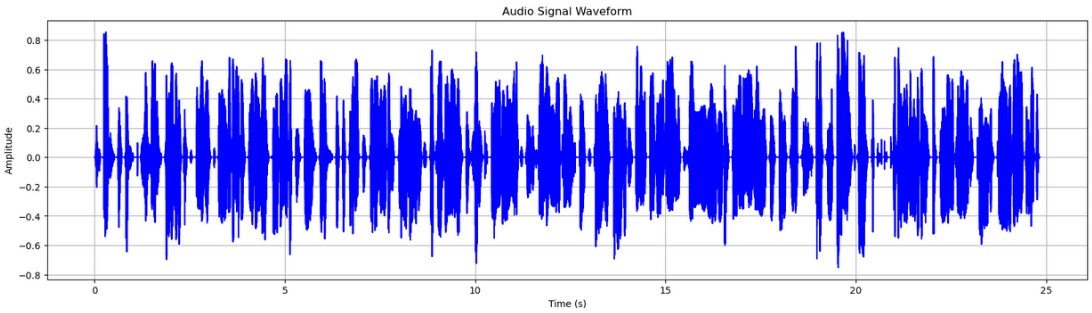


Figure 7.6: Sixth iteration

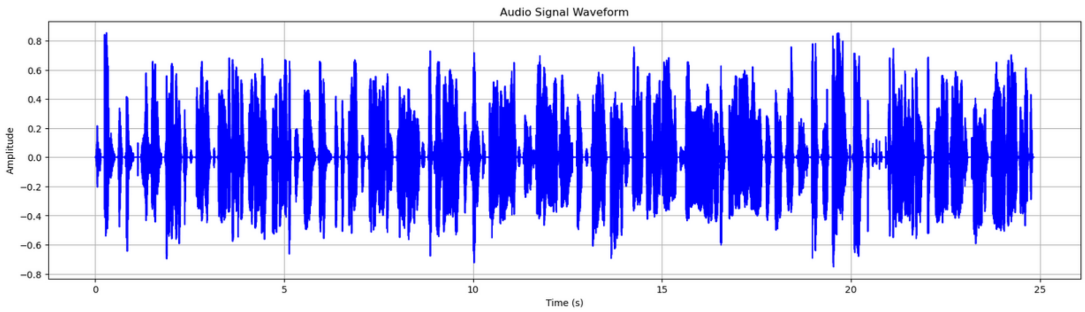


Figure 7.7: Seventh iteration

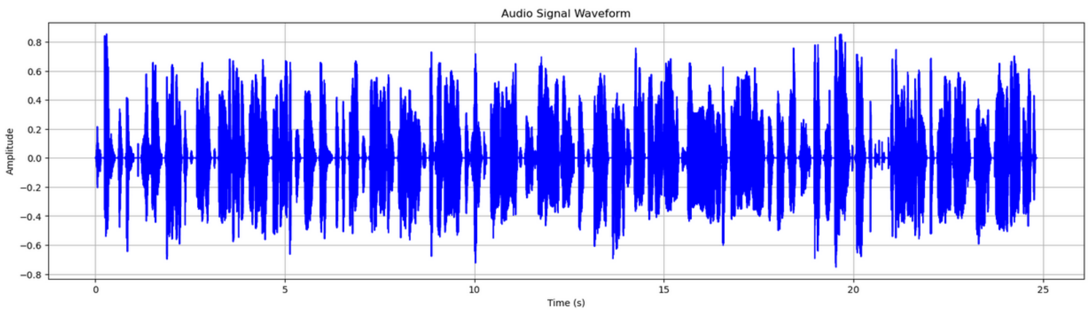


Figure 7.8: Eighth iteration

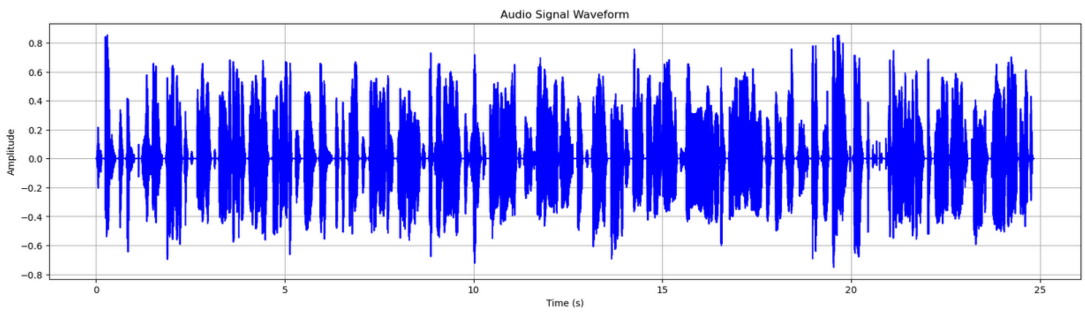
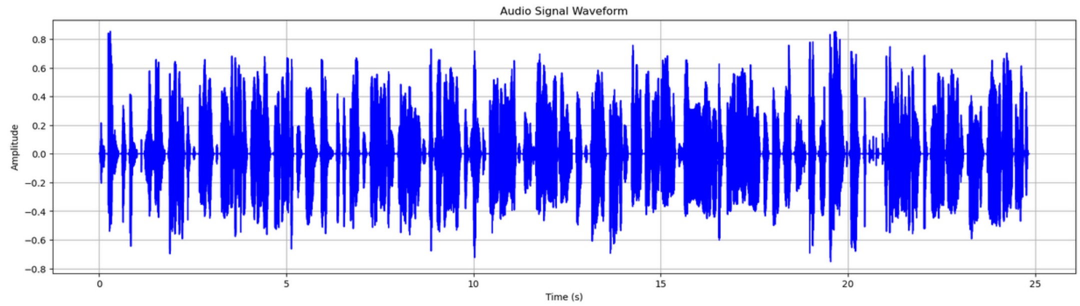


Figure 7.9: Ninth iteration




```
Updated Thresholds iter  10 : [14.00048237  0.07958321  0.91658276]
```

Figure 7.10: Tenth iteration

```
Updated Thresholds iter  1 : [14.00066325  0.07936077  0.92001033]
Updated Thresholds iter  2 : [14.00045471  0.07928772  0.92038466]
Updated Thresholds iter  3 : [14.00028033  0.07932934  0.91988006]
Updated Thresholds iter  4 : [14.00007656  0.08021567  0.91995216]
Updated Thresholds iter  5 : [13.99998978  0.08067156  0.92086317]
Updated Thresholds iter  6 : [13.99993102  0.08038678  0.92032259]
Updated Thresholds iter  7 : [13.99912897  0.08010841  0.92047161]
Updated Thresholds iter  8 : [13.99853441  0.0794859   0.92106533]
Updated Thresholds iter  9 : [13.99824809  0.07986231  0.92020951]
Updated Thresholds iter  10 : [13.99832774  0.07896295  0.91951545]
```

Figure 7.11: Updated parameters

After each iteration, the parameter values are updated. We stop the updates once the values start oscillating, indicating the optimal parameters for that particular speaker is attained.

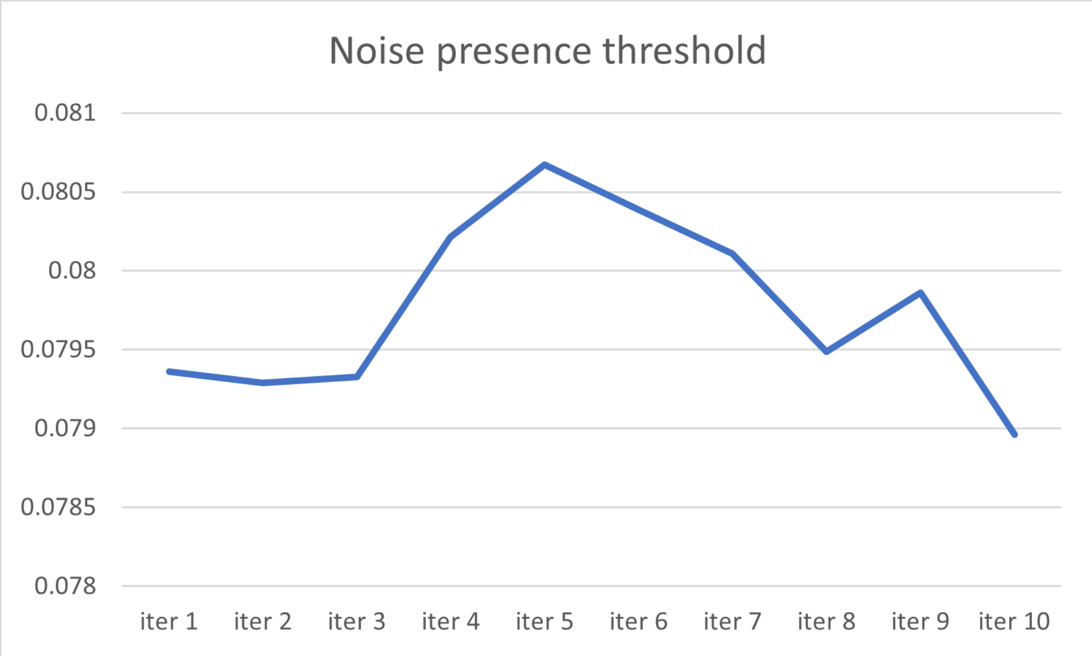


Figure 7.12: Noise presence threshold

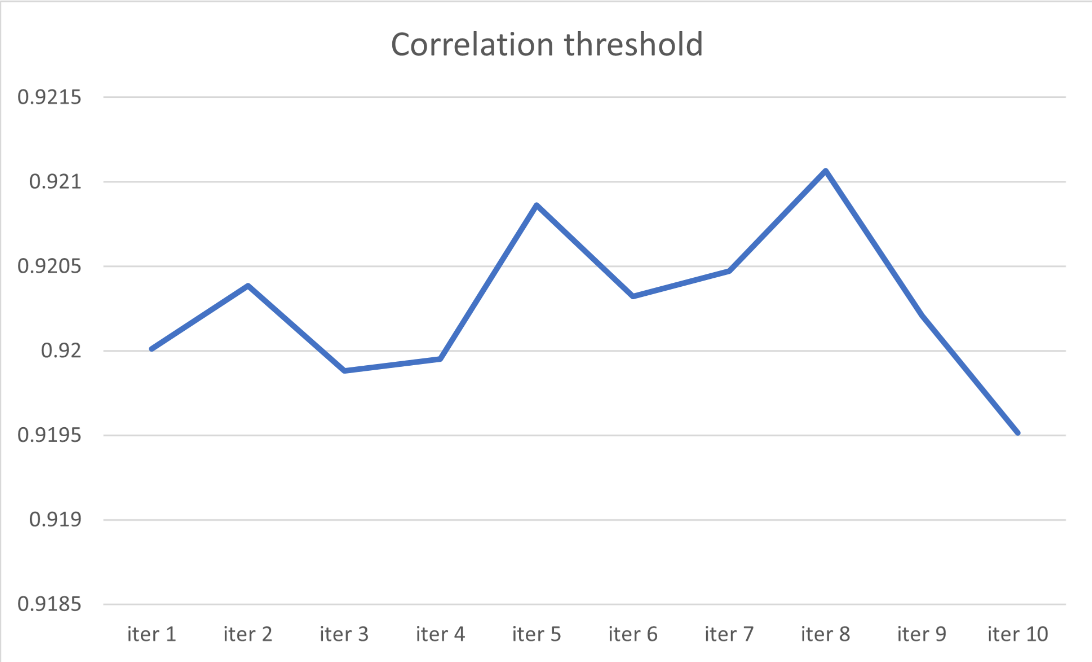


Figure 7.13: Correlation threshold

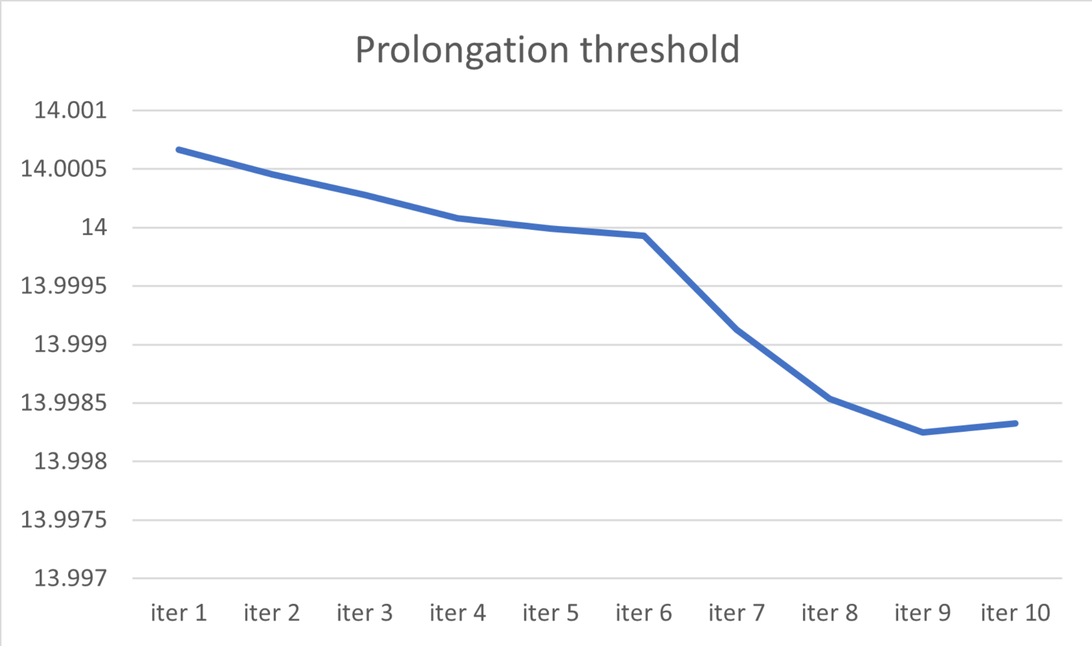


Figure 7.14: Prolongation threshold

The outcomes presented here are derived from our experi- mental investigation, showcasing 10 iterations on a stuttered audio file before and after stutter removal. In order to simulate adaptive stutter correction over time, we performed multiple iterations on the same audiofile. The aim is that in real time, with multiple stuttered audios from a single person, the model will adapt to their speaking style. The graphs show the threshold values obtained from the reptile MAML model. The optimal values for this varies from speaker to speaker and hence depends on the stutter detected from the output audio from the DSP pipeline. We stop the updations after a reasonable number of iterations.

Chapter 8
Future Scope

Our model can be developed to cater to various real-world scenarios, making speech correction more accessible and effective for different user needs. In speech therapy, it can assist individuals with stuttering by providing real-time feedback and correction suggestions. By integrating it into mobile applications or wearable devices, users can practice speaking fluently anytime, without requiring frequent visits to a therapist. Additionally, for educational settings, the model can be adapted to help children with speech delays by providing interactive exercises and personalized training sessions based on their speech patterns.

In professional environments, our model can be used in voice assistants and call centers to enhance speech clarity and reduce communication barriers. Public speakers or individuals preparing for presentations can use the system to refine their speech delivery by analyzing fluency and suggesting corrections. For multilingual applications, expanding the model to recognize and adapt to various accents and dialects will make it valuable for global communication, enabling better speech recognition in diverse linguistic settings. Furthermore, integrating EEG-based brain wave analysis can help individuals with neurological speech disorders, such as Parkinson's disease, by predicting and correcting speech disruptions more accurately. This adaptability makes the model useful across healthcare, education, accessibility tools, and professional communication.

Chapter 9
Conclusion

Adaptive Enhancement of Stuttered Speech Correction is a real-time system designed to address repetitions, long pauses, and prolongations in disfluent speech. By leveraging signal processing techniques such as Mel Frequency Cepstral Coefficients (MFCC) and Linear Predictive Coefficients (LPC), the model achieves high accuracy while maintaining minimal computational demands, making it suitable for deployment on low-power devices. Unlike traditional approaches that focus solely on detection, this system actively corrects speech, ensuring a fluent and natural-sounding output. Its adaptive thresholding mechanism allows it to function independently of language constraints, making it accessible across diverse linguistic backgrounds. By dynamically adjusting to individual disfluency patterns, the system provides an inclusive solution for users with stuttering disorders to interact with voice-enabled devices more effectively. Future advancements could involve expanding the system's capabilities to handle additional types of disfluencies and refining adaptive thresholding techniques to further improve accuracy. With its efficient processing and comprehensive correction abilities, this model presents a promising solution for applications in speech recognition, communication aids, and accessibility technology.

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